COMP3850 Pty Ltd

Feasibility Study

Defensive Team Project (Robust Privacy-Preserving Record Linkage Models)

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1. Introduction

The purpose of this feasibility study is to define the scope of the Privacy-Preserving Record Linkage (PPRL) Defence Project and assess its strategic, economic, technical, operational, and planning feasibility. The project follows a structured implementation process consisting of project planning, model development, evaluation, and final delivery. The supplied PPRL pipeline consists of four stages: Data Pre-processing, Blocking or Filtering, Distance-Aware Data Encoding, and Matching & Linkage. The team will focus on the final stage, **Matching & Linkage**, by implementing four additional machine learning linkage models alongside the supplied threshold-based model and evaluating all five models based on privacy, linkage quality, and computational efficiency.

1.1 Problem Identification

The project aims to address the challenge of securely linking records across multiple organisations while preserving the privacy of sensitive personal information.

The main problems include:

- Protecting sensitive personal information so that identifiers are never exposed in raw form.
- Maintaining high linkage quality despite errors, formatting differences, and corrupted data.
- Improving computational efficiency by reducing unnecessary record comparisons.
- Identifying the linkage model that provides the best balance between privacy, accuracy, and computational efficiency.

1.2 Opportunities

- Improve secure data sharing between organisations without exposing sensitive information.
- Enhance compliance with modern privacy protection principles.
- Increase confidence in privacy-preserving machine learning techniques.
- Improve record linkage accuracy for unstructured and noisy datasets.
- Provide a comparative evaluation of multiple linkage models to identify the most effective solution.
- Develop a reusable framework for future privacy-preserving record linkage research.

1.3 Mandates

- Implement Bloom filter encoding for privacy-preserving record linkage.
- Implement four additional linkage models alongside the supplied threshold-based model.
- Ensure all linkage models incorporate Differential Privacy (DP).
- Evaluate all five models using privacy, linkage quality, and computational efficiency metrics.
- Complete the required COMP3850 project deliverables, including project documentation, prototype, and final presentation.
- Comply with relevant privacy, security, and ethical standards when handling sensitive data.

1.4 Assumptions

- The supplied baseline PPRL code is functional and available for use.
- The provided datasets are sufficient for training and evaluating all linkage models.
- All team members have access to the required software, Python development environment, Jupyter Notebook, GitHub repository, and computing resources.
- Project requirements and resources supplied by the project sponsor will remain available throughout the project.
- The project focuses solely on defensive Privacy-Preserving Record Linkage techniques and does not include attack implementations.
- The project sponsor has provided the necessary datasets, baseline code, and supporting documentation required for development.

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2. Current Situation

The project sponsor, Macquarie University School of Computing, is conducting research into Privacy-Preserving Record Linkage through both defensive and offensive project streams. The team is responsible for the defensive component, which focuses on improving secure record linkage while maintaining privacy.

The sponsor has provided an existing PPRL pipeline consisting of four stages:

- **Data Pre-processing** – Cleans, standardises, and prepares records for processing.
- **Blocking/Filtering** – Reduces the number of record comparisons by grouping similar records.
- **Distance-Aware Data Encoding** – Uses Bloom filters and Differential Privacy to encode sensitive information before comparison.
- **Matching & Linkage** – Compares encoded records and determines whether they represent the same individual.

A baseline threshold-based linkage model has already been provided. The project will extend this implementation by developing four additional linkage models and evaluating all five models to determine the most effective solution for balancing privacy, linkage quality, and computational efficiency.

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3. Benefits

3.1 Tangible Benefits

Improved Linkage Performance

- Development and comparison of five Privacy-Preserving Record Linkage models.
- Improved linkage accuracy through evaluation of multiple machine learning approaches.

Improved Efficiency

- Reduced manual record matching through automated linkage.
- Improved computational performance through efficient matching techniques.

Research Outcomes

- A reusable framework for secure record linkage.
- Comparative evaluation identifying the most effective linkage model.
- Benefits for healthcare organisations, government agencies, and researchers handling sensitive information.

3.2 Intangible Benefits

- Improved protection of sensitive personal information.
- Increased confidence in secure data sharing.
- Better compliance with privacy and ethical requirements.
- Enhanced understanding of privacy-preserving technologies.
- Supports collaboration between organisations without exposing confidential information.

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4. Alternative Solutions

4.1 Alternative 1 – No Action

4.1.1 Advantages

Strategic: Zero effort needed and no schedule risk.

Economic: Zero additional costs to implement.

Technical: No new code to maintain, and no risk of introducing defects into a working baseline.

Operational: Simplifies administration, maintenance, and access management.

Planning: No planning is required.

4.1.2 Disadvantages

Strategic: Fails the mandated project requirements as there is no deliverable.

Economic: Unstructured data remains unlinkable.

Technical: No analysis to check which linkage model suits which party.

Operational: Custodians of the data will have to choose between not linking the data and trusted third party arrangements that concentrate risk.

Planning: Without a schedule there is no outcome to refer to in the future.

4.2 Alternative 2 – Linkage without Privacy Protection

4.2.1 Advantages

Strategic: Highest achievable linkage quality as there is no need to account for encoding loss, noise, or collisions.

Economic: Lowest computation cost, since the encoding and noise stages are absent.

Technical: Simplest to implement as comparison functions apply directly to the raw values without parameters.

Operational: Allows organisations to independently manage their own data while participating in secure record linkage.

Planning: Fastest to deliver.

4.2.2 Disadvantages

Strategic: Does not solve the stated problem as parties are not able to exchange raw identifiers.

Economic: Without privacy, it may face legal problems from affected parties and enforcement.

Technical: Offers no defence whatever against a compromised or adversarial linkage unit.

Operational: Requires a trusted third party and strict data sharing agreements, concentrating risk to a single point of failure.

Planning: Requires rigorous planning on the data privacy policies and practices to mitigate damage.

4.3 Alternative 3 – Bloom Filter Encoding without Provable Guarantees

4.3.1 Advantages

Strategic: Achieves a relatively secure PPRL that is computationally efficient.

Economic: Removes the need for additional computation therefore increasing computational efficiency.

Technical: Achieves privacy and fuzzy matching simultaneously, provides privacy through collisions.

Operational: No party or linkage unit will see raw identifiers.

Planning: Already implemented in the provided source code for structured attributes, so incremental effort is low

4.3.2 Disadvantages

Strategic: Unprovable privacy fails the deliverable requirement.

Economic: May face repercussions in legal matters as a potential breach of privacy.

Technical: Susceptible to cryptanalysis and frequency based attacks, collisions produce false positives.

Operational: Parameter selection demands expertise.

Planning: Requires significant technical planning and testing due to complexity of tools.

4.4 Alternative 4 – Bloom Filter Encoding with Differential Privacy (PPRL Model)

4.4.1 Advantages

Strategic: Supplies mathematically provable guarantees.

Economic: More clients in search of the linkage tool with privacy guarantees.

Technical: It provides explicit control through epsilon.

Operational: With strong privacy, it enables the widest set of collaborations with organisations that have sensitive data.

Planning: Avoids requiring significant legal or ethics planning as there is no leakage of raw data.

4.4.2 Disadvantages

Strategic: Difficult to implement as requires specialised knowledge.

Economic: Additional computation for noise generation.

Technical: Noise reduces linkage quality.

Operational: Difficult for non-specialist staff to configure, maintain and utilise.

Planning: Largest scope of the four alternatives in both implementation and experimentation.

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5. Recommended Solution

5.1 Recommended Solution

Alternative 4, **Bloom Filter with Differential Privacy (Privacy-Preserving Record Linkage)**, is the recommended solution. This approach best supports the project's objective of securely linking records while allowing participating organisations to retain control of their own sensitive data. By combining Bloom filter encoding, differential privacy, and multiple linkage models, the solution provides an effective balance between privacy protection, linkage quality, computational efficiency, and scalability. It also aligns with the project's implementation plan and provides a strong foundation for the comparative evaluation of all five linkage models.